

PROJECT TOPIC: SCHOOL DISASTER EVALUATION PLAN

Topic Description: Comprehensive Disaster Evaluation and Management Plan for a Junior College / School Campus: Hazard Vulnerability, Safety Audits, and National Day Assembly Presentation

Disaster management is an indispensable imperative for educational institutions, where thousands of vulnerable young students, teaching staff, and administrative personnel congregate daily within multi-story buildings. Educational campuses are vulnerable to a wide array of potential natural hazards (such as earthquakes, cloudburst urban flooding, and severe cyclones) and human-induced technological emergencies (including laboratory chemical fires, electrical short-circuits, structural collapse, and stampedes).

This project presents a comprehensive, display-ready **School Disaster Evaluation Plan** for our Junior College campus, developed in strict accordance with the statutory National Disaster Management Authority (NDMA) guidelines and the *National School Safety Policy, 2016*.

The project conducts a thorough structural and non-structural hazard identification audit across all college wings, evaluates vulnerable choke-points, maps emergency evacuation corridors and assembly grounds, designates specific operational roles for the School Disaster Management Committee (SDMC), and details an educational assembly briefing designed to be explained during the National Day campus assembly drill.

2. Importance of the topic (विषयाचे महत्त्व)

Formulating and evaluating an institutional school disaster plan is a vital, lifesaving civic necessity:

Key Dimensions of Institutional Safety Importance :-

- **Preventing Catastrophic School Tragedies:** Past national tragedies (such as the 2004 Kumbakonam school fire and the 2001 Gujarat earthquake) proved that the vast majority of child casualties were caused by locked emergency exits, narrow stairwells, and complete absence of evacuation drills.
- **Mitigating Non-Structural Hazards:** Research shows that over 60% of injuries during earthquakes are caused by non-structural elements—such as heavy unsecured laboratory almirahs falling, shattering window glass, and loose ceiling fans—which can be secured in advance at near-zero cost.
- **Eliminating Mass Panic & Stampedes:** During sudden crises, unguided panic kills more people than the actual hazard. Rehearsed evacuation routes ensure orderly, calm, and swift movement to open ground within 2 to 3 minutes.
- **Empowering Students as Community First Responders:** Training Standard XI students in disaster preparedness instills practical first-aid, firefighting, and leadership skills that protect surrounding neighborhoods during regional crises.

A proactive disaster plan transforms a school from an emergency hazard into a resilient community shelter.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The institutional safety investigation was guided by structured disaster management objectives:

Structured Project Objectives :-

- • **Campus Hazard & Vulnerability Audit:** To conduct a detailed physical inspection identifying structural risks (cracks, narrow staircases) and non-structural hazards (chemical storage, loose almirahs) across college buildings.
- • **Evacuation Egress Mapping:** To identify, measure, and map unobstructed primary and secondary evacuation corridors leading from all classrooms and laboratories to the open central playground.
- • **Disaster Management Committee Structuring:** To formulate the organizational chart and operational Standard Operating Procedures (SOPs) for the student and faculty School Disaster Management Committee (SDMC).
- • **Resource & Emergency Kit Inventory:** To audit campus emergency equipment (fire extinguishers, first-aid boxes, siren systems, stretcher access) and verify their operational readiness.
- • **National Day Assembly Presentation Design:** To draft a concise, clear 5-minute educational speech and presentation to explain the disaster plan to all students during the National Day assembly.

The scope encompasses physical campus walk-throughs, building blueprint reviews, and NDMA guideline compliance checks.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized institutional safety audit and hazard evaluation methodology was deployed across the campus:

Methodological Audit Protocol :-

- **Phase 1 – Campus Walk-Through & Vulnerability Mapping:** Conducted a room-by-room safety walk-through across all three floors, checking chemistry/physics laboratories, electrical meter rooms, and stairwells with a standard NDMA checklist.
- **Phase 2 – Non-Structural Hazard Identification:** Documented all heavy furniture, glass panes, overhead water tanks, and wall-mounted apparatus capable of toppling during seismic tremors.
- **Phase 3 – Evacuation Corridor Timing & Measurement:** Measured staircase widths, exit gate clearances, and conducted timed trial walks to calculate the average evacuation time required to clear all classrooms.
- **Phase 4 – Institutional Role Allocation:** Structured four dedicated volunteer student response teams: Warning Team, Evacuation Team, Search & Rescue Team, and First-Aid Team.
- **Phase 5 – Synthesis & Curricular Alignment:** Synthesized findings according to Chapter 7 of the Maharashtra State Board Standard XI EVS project curriculum.

All emergency procedures were designed in compliance with the National School Safety Policy.

5. Observations (निरीक्षणे)

The systematic campus safety audit recorded key vulnerabilities, existing safety assets, and functional emergency plans:

5.1 Campus Hazard Identification and Safety Audit Matrix

Campus Area / Wing	Identified Hazard / Vulnerability	Existing Mitigation Asset	Recommended Corrective Action
Main Classroom Building (3 Floors, 24 Classrooms)	Doors open inward; narrow central staircase causes bottleneck during rapid descent.	Two wide exterior fire escape stairs available at east and west ends.	Re-hang classroom doors to swing outwards into corridors; designate separate 'Up' and 'Down' staircases.
Chemistry Laboratory (Ground Floor)	Glass bottles of volatile acids (HCL, H ₂ SO ₄) stored on open high shelves; LPG cylinders.	Two 5-kg ABC dry powder fire extinguishers mounted near doorway.	Install guard-rails on reagent shelves; relocate acid bottles to low floor-level trays; service extinguishers bi-annually.
Electrical Meter Room (Under Main Staircase)	Cluttered with discarded wooden desks; high risk of electrical short-circuit fire blocking stairs.	Main circuit breakers (MCBs) installed; sand buckets present.	Immediately remove all combustible wooden junk; relocate electrical main board away from primary evacuation route.
College Assembly Playground	Clear, open ground situated 25 meters away from building fall-zones; wide exit gates.	Designated Primary Safe Assembly Area (Capacity: 1,500 students).	Paint clearly marked class-wise muster zones (XI-A, XI-B) with white lime powder for rapid headcount roll calls.

5.2 SDMC Student Volunteer Teams Formed

- Warning Team: Activates alarm bell (3 long rings); announces calm evacuation instructions over PA system

6. Analysis of data (माहितीचे विश्लेषण)

Analytical examination of the campus safety audit demonstrates that institutional disaster management is an ongoing operational culture:

Core Analytical Inferences :-

- **The Lethal Danger of Inward-Opening Doors:** The audit revealed that most classroom doors swing inward. In a panic rush, pushing students trap the door shut against the frame, creating fatal crushes. Re-hanging doors to swing outward in the direction of exit travel is an urgent, non-negotiable safety priority.
- **Non-Structural Mitigation as Low-Cost Safety:** Over 75% of identified classroom risks—such as unanchored almirahs and wall-mounted projectors—can be eliminated simply by bolting them securely to masonry walls with L-brackets, requiring minimal financial investment.
- **The 'Drop, Cover, and Hold' Reflex:** During an earthquake, attempting to run down stairs while the ground is shaking leads to severe falls and injury. The plan mandates that students immediately practice *Drop, Cover, and Hold* under sturdy desks for 60 seconds until tremors cease before commencing orderly evacuation.
- **National Day Drill as an Educational Catalyst:** Briefing the entire student body during the National Day assembly ensures that disaster preparedness is treated as an active civic duty, replacing casual indifference with serious, life-saving discipline.

The analysis affirms that emergency response success depends entirely on pre-rehearsed muscle memory.

7. Results and Conclusions (निष्कर्ष)

Conclusion: A well-evaluated, rehearsed School Disaster Management Plan is the ultimate safeguard protecting young lives against unexpected catastrophes. Transforming our campus into a disaster-resilient institution requires mandatory term drills, non-structural hazard bolting, and vigilant student volunteer leadership.

Action Checklist for Campus Safety Stewardship :-

- ❑ Conduct unannounced campus-wide evacuation fire drills at least twice every academic year.
- ❑ Display floor-wise color-coded evacuation maps in every classroom, laboratory, and hallway.
- ❑ Ensure all emergency exit doors and gates remain permanently unlocked and unobstructed during school hours.
- ❑ Secure all high chemical shelves, almirahs, and library racks to walls with structural steel brackets.
- ❑ Maintain certified, fully stocked First-Aid Kits in every science laboratory and sports room.

8. Bibliography / References (संदर्भ सूची)

1. National Disaster Management Authority (NDMA). (2016). *National School Safety Policy Guidelines*. New Delhi.
2. Maharashtra State Disaster Management Authority (MSDMA). *Standard Operating Procedures for Educational Campuses*. Mumbai.
3. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.
4. Central Board of Secondary Education (CBSE). (2004). *Disaster Management in Schools: Resource Book*. Delhi.